FreshWay Board archive — C05 · Energy and grid capacity

Company total: 115 GWh/year — electricity 76.3 · gas 29.5 · road diesel 9.6. **Energy cost €16.3 mln = 0.74% of net revenue**, at €0.19/kWh and €0.60/m³. Stores 88.5 GWh (380 kWh per m² sales area); DCs and EFC 15.8 GWh. **Refrigeration is ~73% of store electricity and ~69% of DC electricity.** On-site generation 16.8 GWh = 22% of electricity demand.

Site	Electricity/year	Average load	Peak	Contracted	Headroom
CDC Apeldoorn	4.65 GWh	531 kW	~1.06 MW	**1.5 MVA**	**~0.45 MVA**
RDC Noord / Zuid	1.58 GWh each	180 kW	~0.36 MW	0.6 MVA each	~0.25 MVA
EFC	2.62 GWh	300 kW	~0.60 MW	1.5 MVA (incl. van charging)	~0.25 MVA
Hub	0.12 GWh each	14 kW	~30 kW	150 kVA each	
Store (1,400 m²)	531 MWh	61 kW	~121 kW	160 kVA (XL 250)	~40 kVA

The binding constraint, and it is two constraints. Fully electrifying the truck fleet needs about **1.2 MW** of charging (24 chargers × 50 kW). Against 0.45 MVA of headroom that is a **shortfall of some 1.3 MVA**, and a heavier connection takes one to two years — up to ten in a congested area.

The second constraint is the night itself, and it is a scenario, not a measurement. The charging window runs 18:00-06:00: twelve hours. What the chain actually draws in that window **today** is in the meter data, and it is only a part of those twelve hours — because **8 of our 54 tractors are electric**. What the window looks like at 54 is a calculation on our own kilometres and our own meter, and nobody here has made it. **Do not quote a full-electrification figure as if it were a reading**; make the sum, show the assumption underneath it, and say which of the two you are reporting. The van fleet is the easy half: **82 vans**, 3.4 MWh a night at 0.90 MW, charging in 3.8 hours and fitting comfortably. *Still to be set: contract end dates and the status of any connection request. Storage receives this file on day one; it is the basis of the Energy & Capacity File.*

> **You have this card from day one, and you are the only department that does.** Storage holds the FreshWay Energy & Capacity File for the whole company and cannot maintain it without these figures. Every other department has to file an Information Request to see them — which means they will come to you. Do not forward the card; answer the question.

> **GenAI transparency.** This card was written with Claude (Anthropic) by the course team, based on the FreshWay case design and a source-checked figure sheet. No figures were generated by the model. Human-AI Agency Label: **Co-Creator**. Label source: Boetje, J., & Baake, G. (2026). *Nine prototypical human-AI agency patterns* [Figure]. figshare. https://doi.org/10.6084/m9.figshare.31706884 (CC BY 4.0).

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